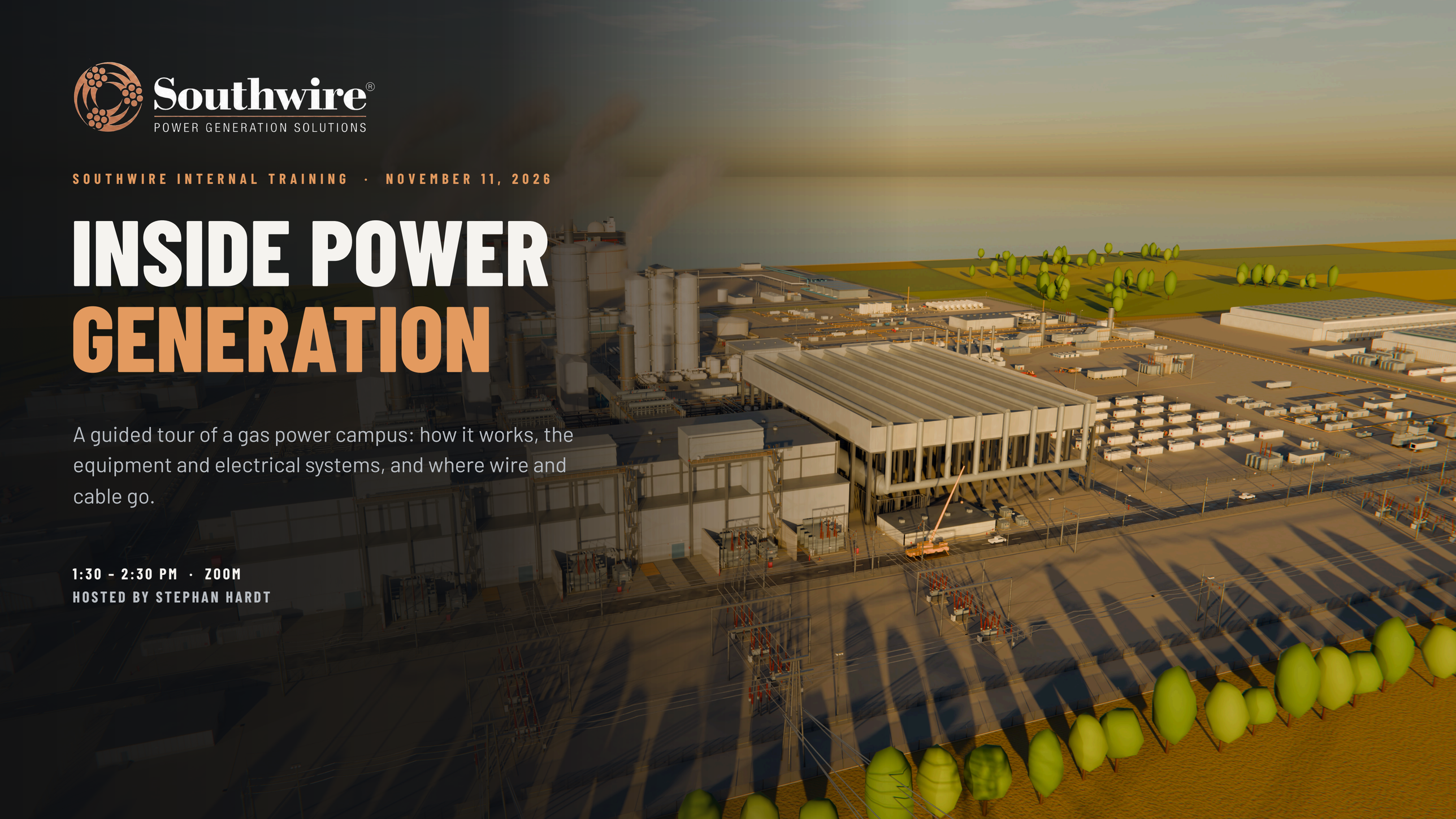
Southwire®
POWER GENERATION SOLUTIONS

SOUTHWIRE INTERNAL TRAINING · NOVEMBER 11, 2026

INSIDE POWER GENERATION

A guided tour of a gas power campus: how it works, the equipment and electrical systems, and where wire and cable go.

1:30 - 2:30 PM · ZOOM
HOSTED BY STEPHAN HARDT



TODAY · 60 MINUTES

THE HOUR

1:30	**WELCOME** why this matters	2 MIN
1:32	**MARKET AND FUNDAMENTALS** how power is made, the gas build; the machines, the grid, cable, who buys	28 MIN
2:00	**THE 3D TOUR** seven stops: what it does, its electrical system, its cable	18 MIN
2:18	**WIRE AND CABLE TAKEAWAYS** the whole plant in one number, products, pain points	6 MIN
2:24	**Q&A** questions from the chat	6 MIN

No engineering background needed. At every stop you get three things: **what it does**, its **electrical system**, and **the cable**.

Questions in the chat; we answer them at the end.

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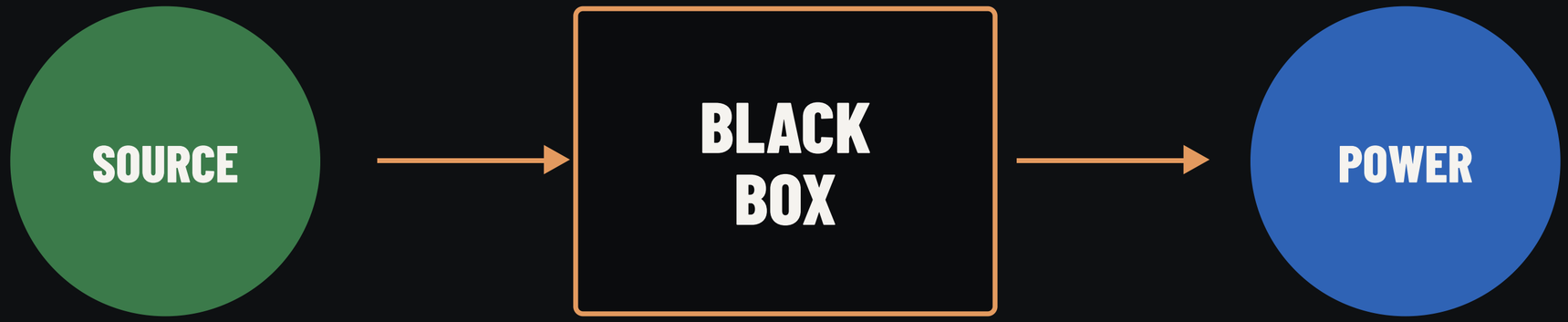
EVERY POWER PLANT IS THIS

Input → conversion → output. The complexity is what's inside the box, and what's around it.



INPUT

The energy that goes in: fuel (gas, coal, uranium, biomass) or nature (sun, wind, water).

CONVERSION

The machine that turns the source into electricity: a turbine and generator, an engine, a solar panel, a fuel cell.

OUTPUT

Electricity out: to the grid through a switchyard, or straight to a load behind the meter.

Every plant, from a gas turbine to a solar farm, fits this picture. **What changes is the source and the machine in the box.** Next: the four families of power generation, then we open the box of a gas plant.

HOW ELECTRICITY IS MADE: FOUR FAMILIES

THERMAL

Heat spins a turbine and a generator.

- **NATURAL GAS** — FIRM
  simple cycle · combined cycle · engines · aero ← today
- **COAL** — FIRM
- **NUCLEAR** — FIRM
- **BIOMASS AND WASTE** — FIRM
- **GEOTHERMAL** — FIRM
- **OIL / DIESEL** — FIRM

WATER AND WIND

Moving water or air spins the turbine.

- **HYDRO** — FIRM*
- **WIND, ONSHORE AND OFFSHORE** — VARIABLE

Not on our campus; Southwire cables them too: hydro, onshore and offshore wind.

DIRECT

No turbine: sunlight or chemistry makes DC.

- **SOLAR PV** — VARIABLE
- **FUEL CELLS** — FIRM

In SW-3X1: the fuel-cell bank in the modular yard.

STORAGE

Shifts power in time; it does not make it.

- **BATTERIES (BESS)** — HOURS
- **PUMPED HYDRO** — HOURS

In SW-3X1: the battery (BESS) yard.

GAS, ENGINES, AERODERIVATIVES AND PACKAGE BOILERS: ALL THERMAL, ALL ON OUR CAMPUS

> **Almost all of them spin a magnet inside a coil: a generator.** Solar PV, fuel cells and batteries make DC and need inverters. Firm sources run on demand; variable ones follow the weather. Every one of them ends in cable.

Firm = can run on demand. *Hydro depends on water and season. Storage runs for hours, then recharges.

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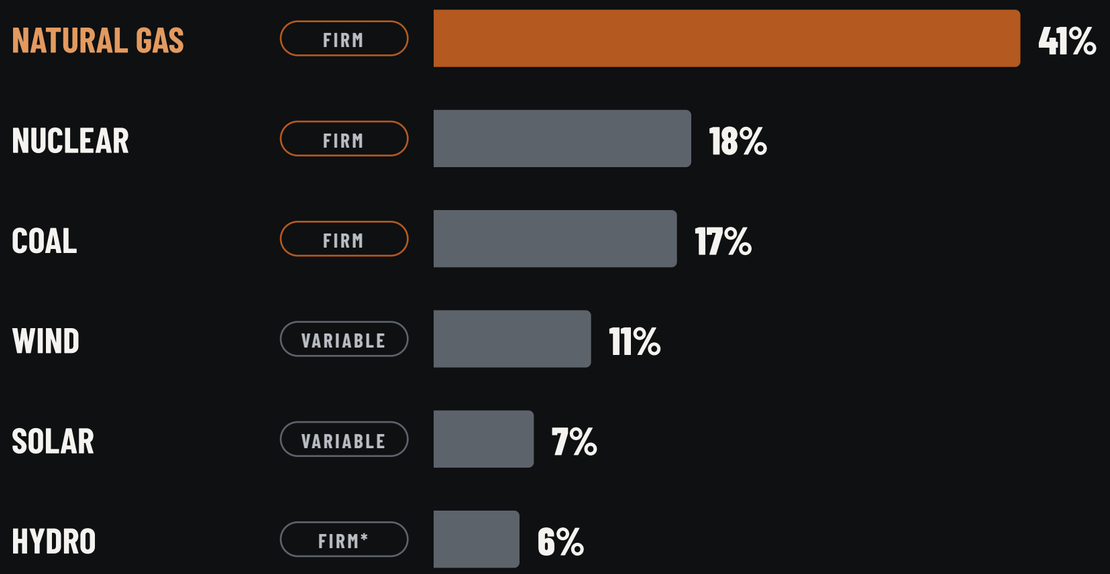
THE MIX TODAY, AND WHY WE ZOOM IN ON GAS

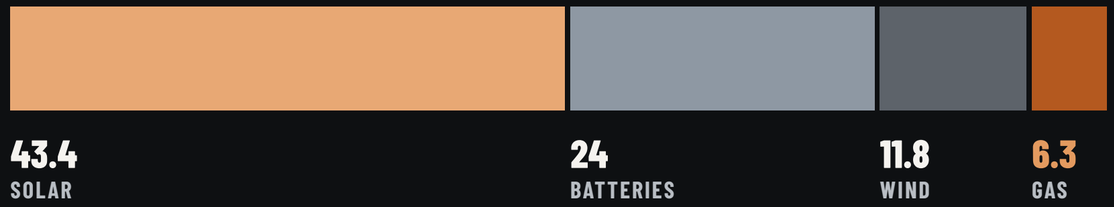
SHARE OF US ELECTRICITY, 2025



Firm = can run on demand. *Hydro depends on water and season. Batteries store power; they do not make it.

NEW CAPACITY PLANNED FOR 2026 · 86 GW

Most of what gets added is solar and batteries. Gas is a small share of the GW, and the one that runs on demand.



FIRM

Gas runs when it is called: day or night, any weather. Wind and solar follow the weather; batteries cover hours, not days.

THE BACKBONE

41% of US electricity today, and the backup that lets more wind and solar onto the grid.

THE NEXT WAVE

Data centres need firm power around the clock: 378 GW of gas is in development, much of it built later this decade.

Southwire cables every one of these: solar, wind, storage, transmission and distribution. **Today we focus on gas**: firm power, a boom-scale build, and some of the most cable-dense plants on the grid.

EIA, Electricity explained (2025 utility-scale generation, rounded) and Today in Energy, Feb 2026 (planned 2026 additions; developers' plans, not all will be built).

A BOOM-SCALE GAS BUILD, AND CABLE IS ON THE CRITICAL PATH

THE MARKET

378 GW

US gas power in development; 52 GW already under construction

THE CABLE

~$1.2B

core wire & cable, 2026–2035, at $8–13M per GW

THE PIPELINE

386

gas projects we track in 40 states, every owner type

THE TRACTION

$50M+

booked in Q3; $30M in September alone

TODAY'S GOAL: KNOW THE PLANT, SO YOU CAN FOLLOW THE CABLE THROUGH IT

Global Energy Monitor, Aug 2026; Southwire PowerGen planning model and internal figures, 2026.

MARKET CONTEXT

LOAD GROWTH IS PULLING GAS POWER FORWARD

166 GW

US peak load growth forecast by 2030

about 90 GW of it from data centres

378 GW

US gas power in development, mid-2026

189 GW tied to data centres

52 GW

under construction, up 76% in H1 2026

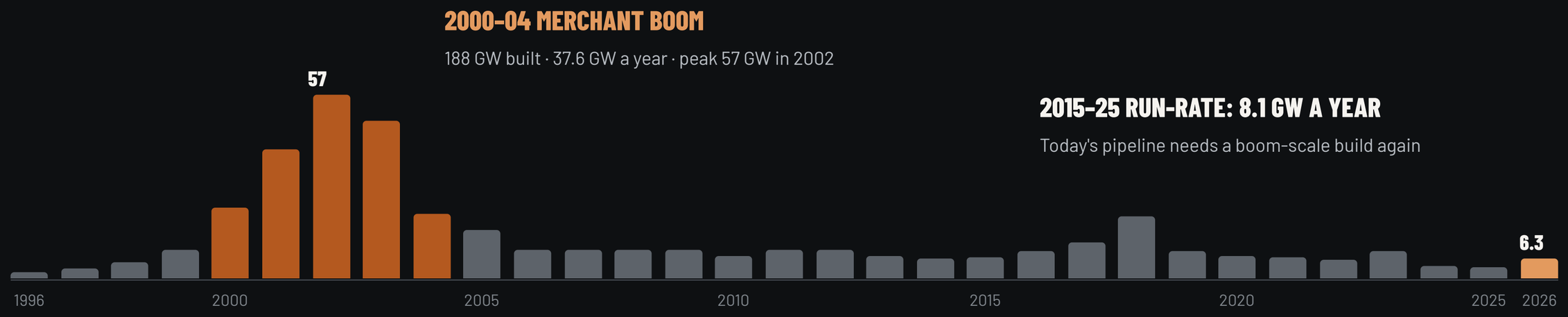
the largest gas build in the world

Up to 7 yrs

quoted lead time for some large gas turbines

procurement decisions are being made now

US GAS CAPACITY ADDED PER YEAR, GW · PROVISIONAL



Grid Strategies, 2025; Global Energy Monitor, Aug 2026; S&P Global, May 2025. History: EIA and Southwire provisional series, to be verified.

OEM ORDER BOOKS

TURBINE BACKLOGS ARE FULL

The three large-frame turbine OEMs together hold **about 220 GW** of gas turbines in backlog and slot reservations. Modular and packaged gas power is growing beside them.

MODULAR

+72%

power-generation sales at the largest engine-generator maker, Q2 2026

BEHIND THE METER

2.3 GW

modular gas fleet for Texas AI data centres, plus 1 GW for another campus

PACKAGED

500 MW

standard modular gas-turbine plant, offered as a product

OEM releases, Q2–Q3 2026 (firm backlog plus slot reservations; gas-turbine backlog; large-frame backlog); Utility Dive, Aug 2026; POWER, Oct 2025 and Feb 2026.

SIZING THE OPPORTUNITY · CONSERVATIVE PLANNING CASE

~$1.2B OF CORE GAS CABLE, 2026–2035

CAPACITY

93.7 GW

modeled additions through 2036, from the 378 GW pipeline: 90% of the 52 GW building, 13.5% of the rest

PLANT INVESTMENT

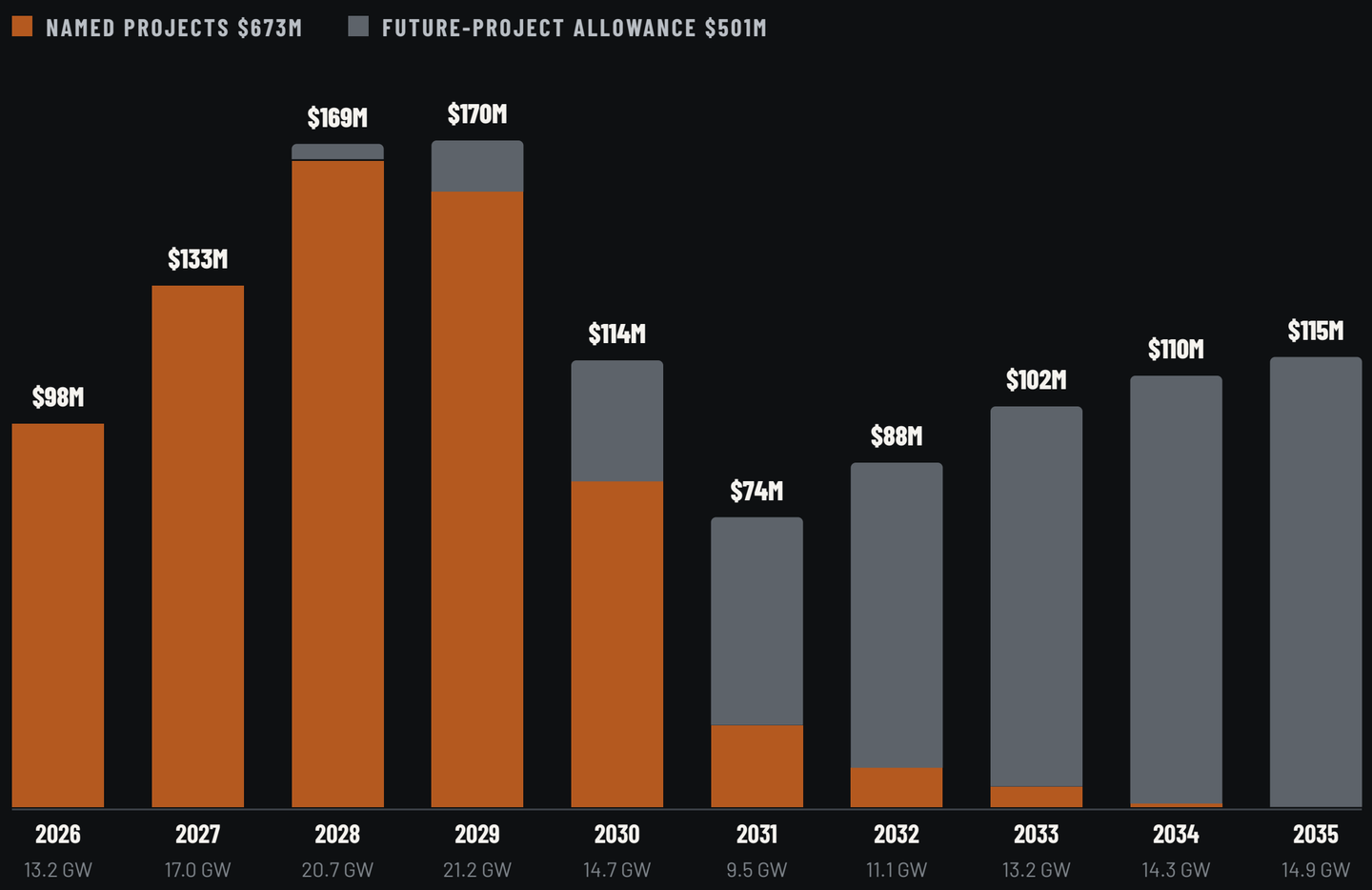
~$180B

93.7 GW × $1,923/kW blended (combined cycle, peakers, engines)

WIRE & CABLE

$0.75–1.2B

$8–13M per GW: under 1% of plant capex, on the critical path to first power

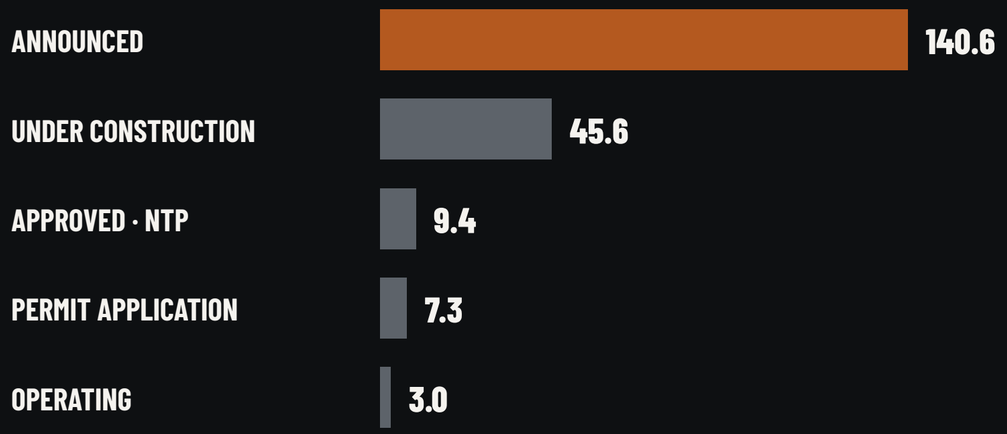


Core cable by purchase-order year; below each year, modeled PO-year GW (not installed GW). GW × $/kW × addressable share × win probability. OEM internal wiring and HV excluded.

Global Energy Monitor, Aug 2026; completion rates, blended cost and cable rates are Southwire planning assumptions.

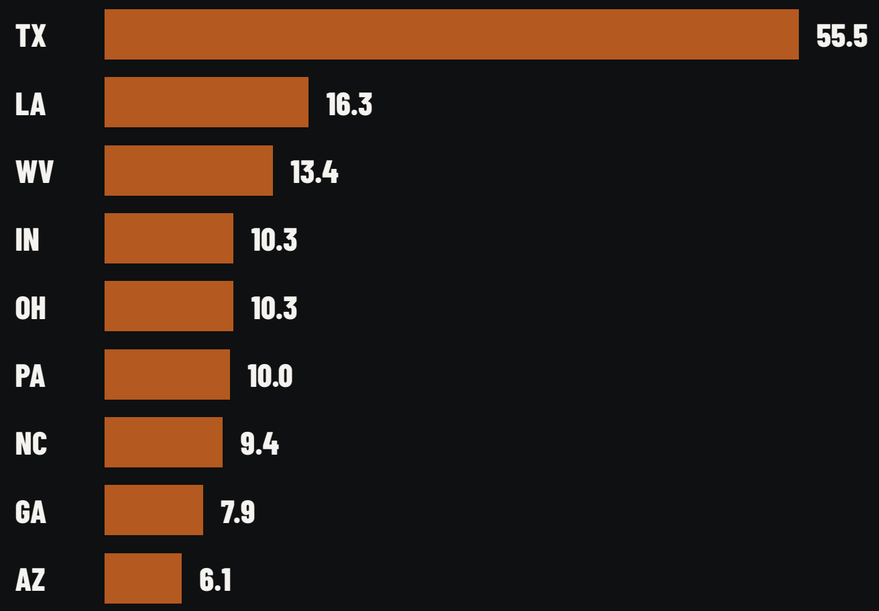
205.8 GW ELIGIBLE, MOST OF IT STILL EARLY

BY STATUS · GROSS GW, COD 2026-2035



68% **of eligible capacity is only announced.** Being on the register does not mean a cable package is open; it means a conversation to start.

TOP STATES · GROSS GW, 386 PROJECTS



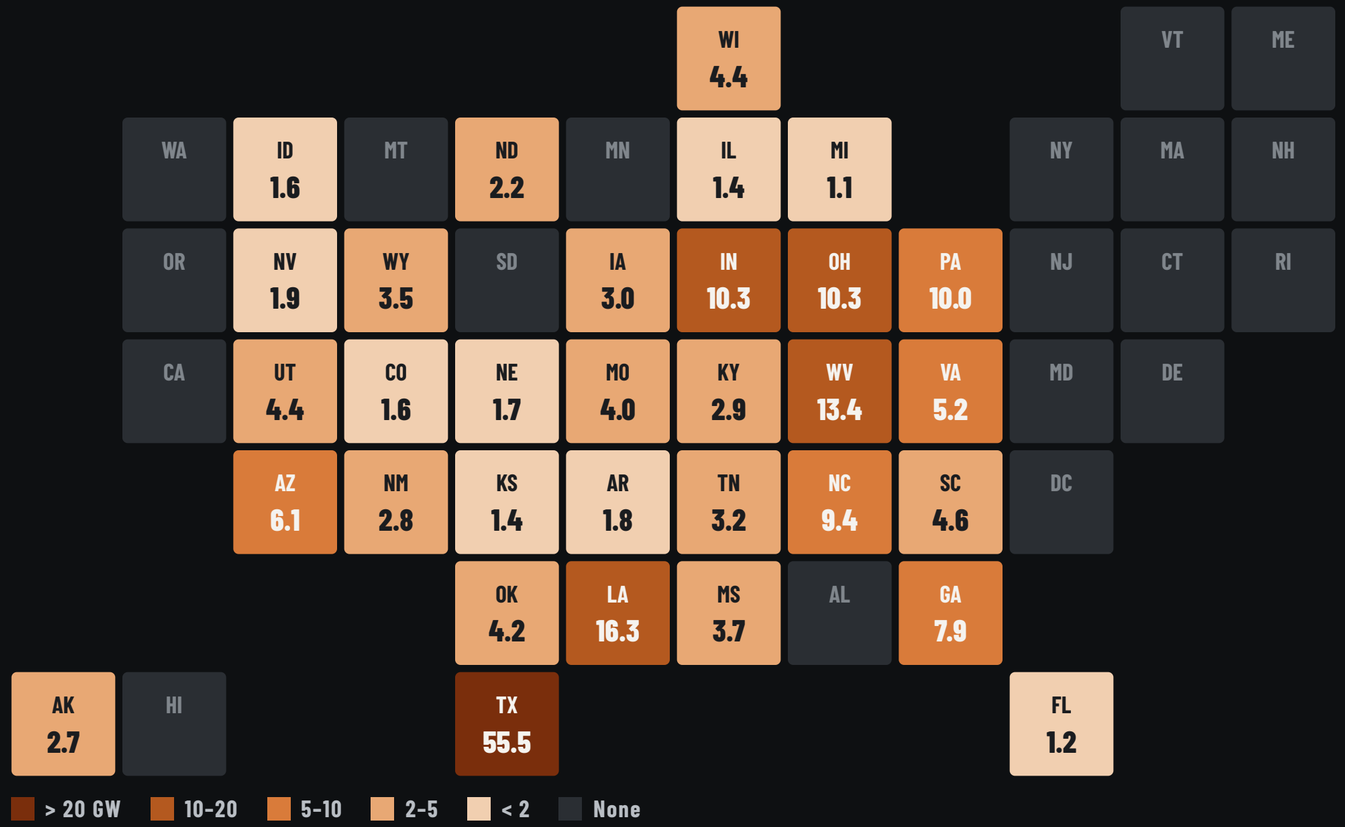
Greenfield 137 GW, brownfield 69 GW. Utilities reuse old sites (coal-site repowers, expansions); independent and behind-the-meter developers mostly build new, often on data-centre campuses.

Southwire PowerGen register, Sep 11, 2026: 386 records across 305 site groups; gross GW, unweighted.

PROJECT CONCENTRATION

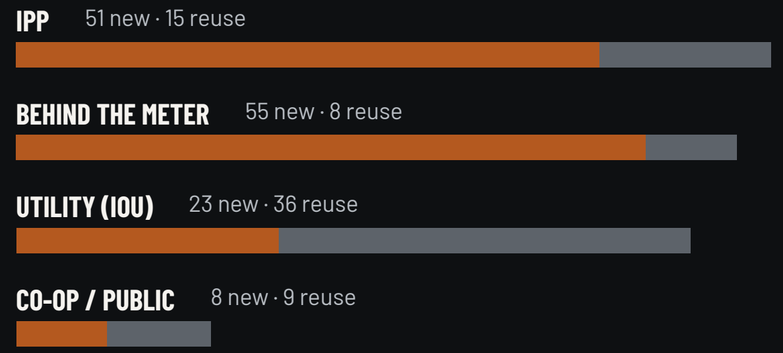
WHERE THE BUILD IS, WHO OWNS IT, WHAT KIND OF SITE

GROSS GW BY STATE · 386 PROJECTS, COD 2026-2035



OWNER × SITE · GROSS GW

GREENFIELD 137 · BROWNFIELD 69



UTILITIES REUSE OLD SITES; IPPS AND BTM BUILD NEW

Greenfield, grid-connected	77.0
Data-centre / BTM campus	54.5
Coal-site repower	41.3
Expansion at an operating plant	21.4
Industrial captive and site reuse	11.6

Southwire PowerGen register, Sep 11, 2026. Gross GW, unweighted. Site classes provisional (Rev01).

OPEN THE BOX: THE PHYSICAL ANATOMY

INPUT
fuel / energy source

BLACK BOX
thermal conversion

OUTPUT
dispatchable power, delivered

0 FUEL INFRASTRUCTURE

- Gas production, pipelines, compression
- Storage and LNG logistics
- Plant-gate fuel delivery

1 FUEL INPUT

- Natural gas, coal, oil / diesel
- Hydrogen blends
- Biomass where it applies

2 THERMAL CONVERSION

- Frame and aero gas turbines, engines (RICE)
- Combined cycle: GT + HRSG + ST
- Modular and H2-ready packages

3 ELECTRICITY OUTPUT

- Dispatchable AC power
- Stable, reliable load
- Baseload, mid-merit or peaking

4 DELIVERY & FLEXIBILITY

- Transmission and distribution
- MV / LV delivery, behind-the-meter on site
- Storage and flexibility support

5 END USE / OFF-TAKE

- Utilities, IPPs, industrial loads
- Data centres and other off-takers
- Critical infrastructure

Back to the black box, now for gas. Gas turbine, combined cycle or engines: **the same physics on every project.** What changes is the machine in the box, and who owns it.

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FOUR WAYS TO BURN GAS FOR POWER

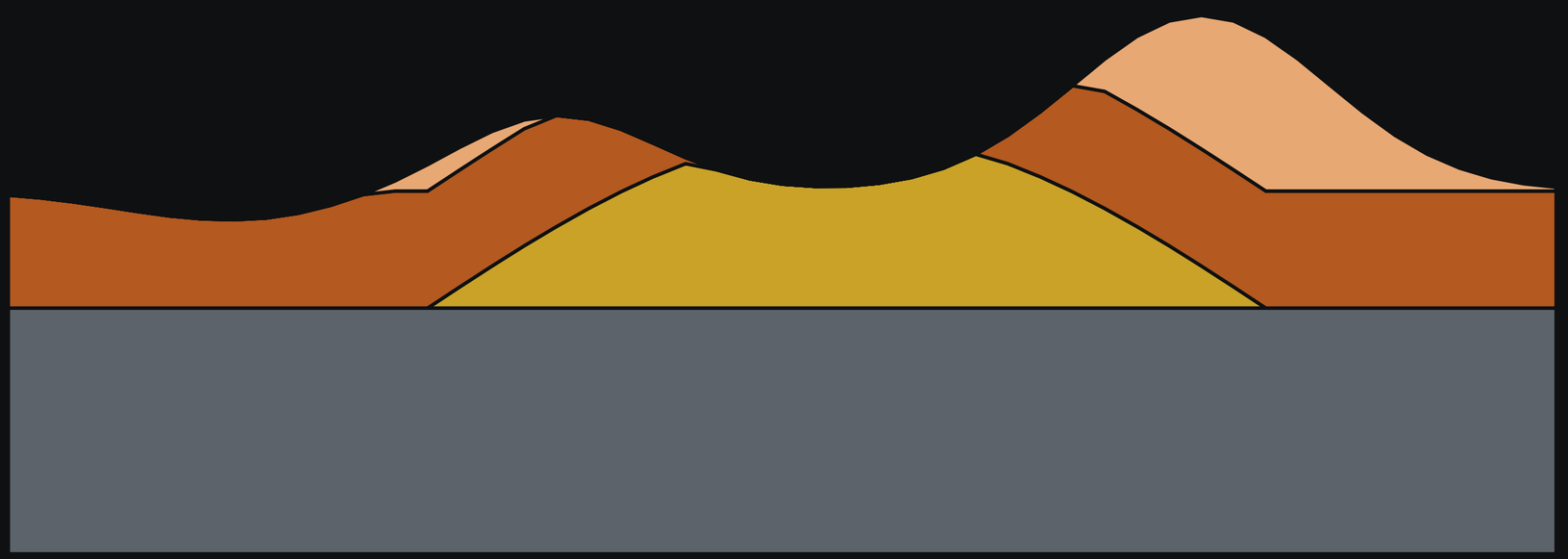
	SIMPLE CYCLE THE PEAKER	COMBINED CYCLE THE WORKHORSE	ENGINES (RICE) THE SPRINTER	AERODERIVATIVE THE JET
UNIT SIZE	**50–400 MW per gas turbine**	**300–1,500+ MW per block**	**2–20 MW per engine, in banks**	**30–60 MW per unit**
EFFICIENCY	**~35–40%**	**~55–64%**	**~40–50%**	**~38–42%**
START TO FULL LOAD	**10–30 min**	**about 1–3 hours**	**about 5 min**	**5–10 min**
RUNS AS	Peaking, mid-merit	Baseload, mid-merit	Peaking, data-centre firming	Peaking, fast backup
IN THE MODEL	A gas turbine alone: exhaust goes straight up the stack	Gas turbines + HRSGs + steam turbine: the SW-3X1 power block	Big gas engines in a hall: the modular yard	A jet engine on the ground: the modular yard

Units that stick: 1 GW = 1,000 MW = 1,000,000 kW. **MW** is power (how hard it runs right now); **MWh** is energy (power × hours). Efficiency is how much of the fuel's energy leaves as electricity; the rest goes up the stack as heat.

Typical ranges for new US units; efficiency on a lower-heating-value basis. Starts vary with design and how hot the unit is.

A DAY ON THE GRID: WHO RUNS WHEN

Demand rises in the morning and peaks in the evening. Each kind of plant takes a layer.



- Peaking: simple cycle, engines, batteries
- Mid-merit: combined cycle
- Solar
- Baseload: nuclear, coal, combined cycle

The evening peak arrives as the sun sets: that is when gas and batteries earn their keep.

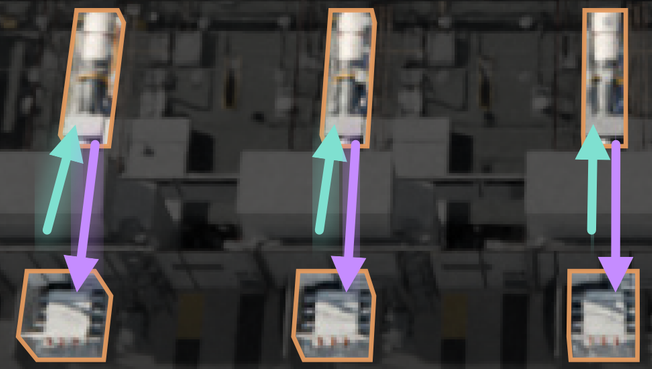
Illustrative summer weekday, not a real grid's data.

Every start and stop cycles the plant, and its cable: heat, vibration and terminations that move. Peakers may start every day.

AIR POWER

THE GAS TURBINE: A JET ENGINE ON THE GROUND

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE

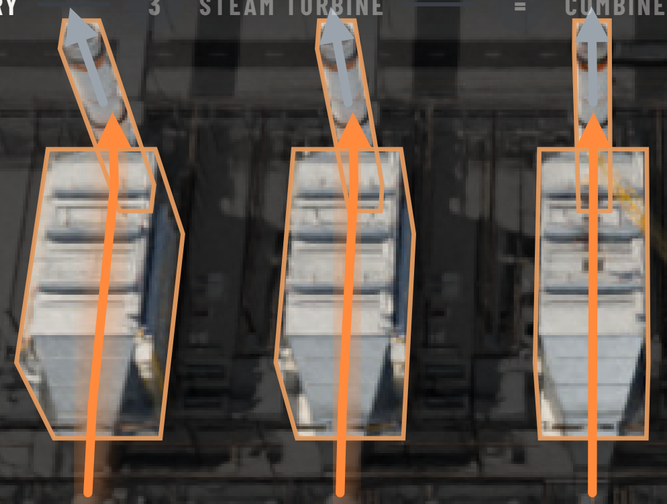


It burns gas in compressed air; the hot gas spins the turbine and its generator. That makes **about two-thirds** of the plant's power, and leaves exhaust at about 1,100 °F.

THE HRSG: CATCH THE HEAT

HOT EXHAUST — TO THE STACK

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE

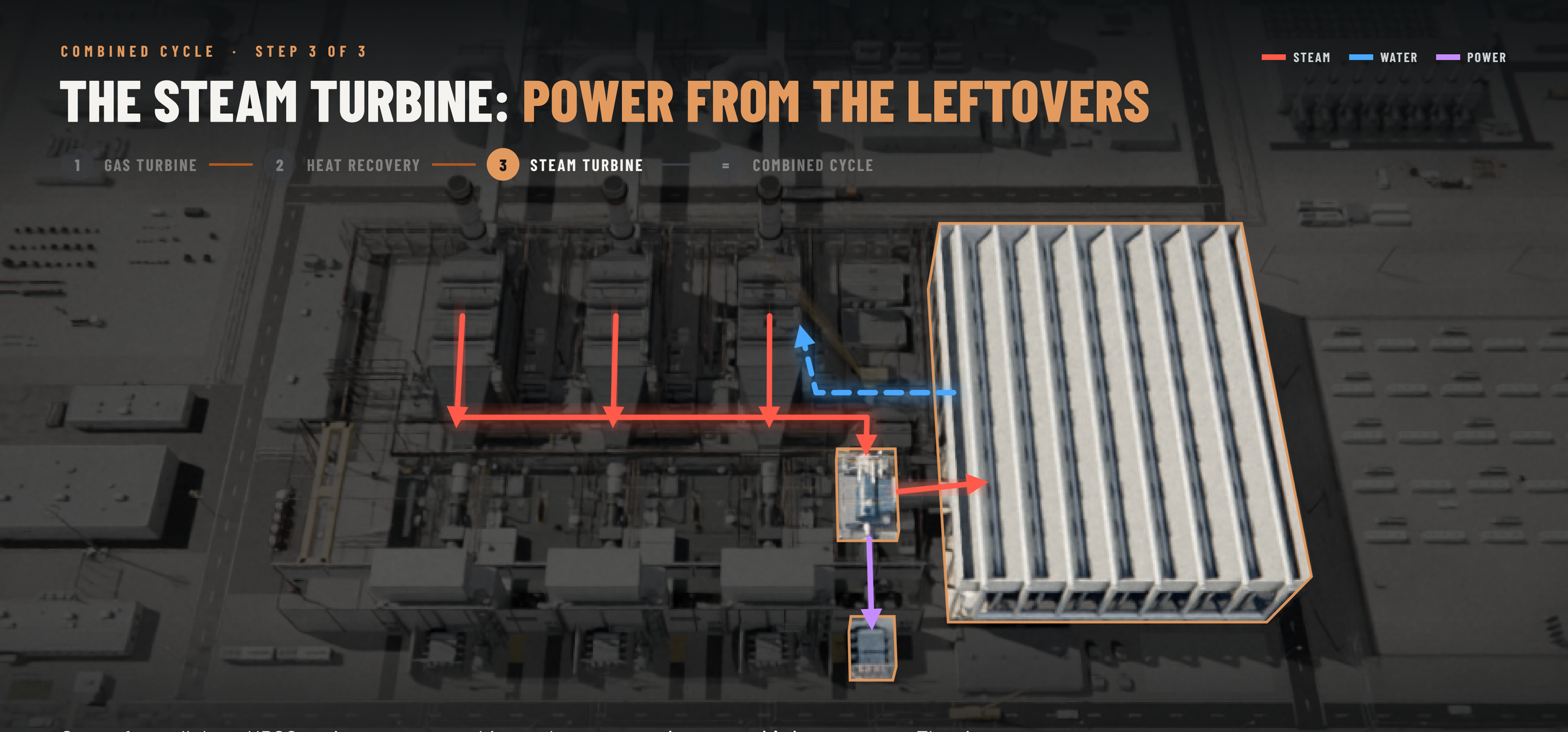


Instead of going up the stack, the hot exhaust runs through a heat-recovery steam generator and boils water into steam. **No extra fuel.**

STEAM WATER POWER

THE STEAM TURBINE: POWER FROM THE LEFTOVERS

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



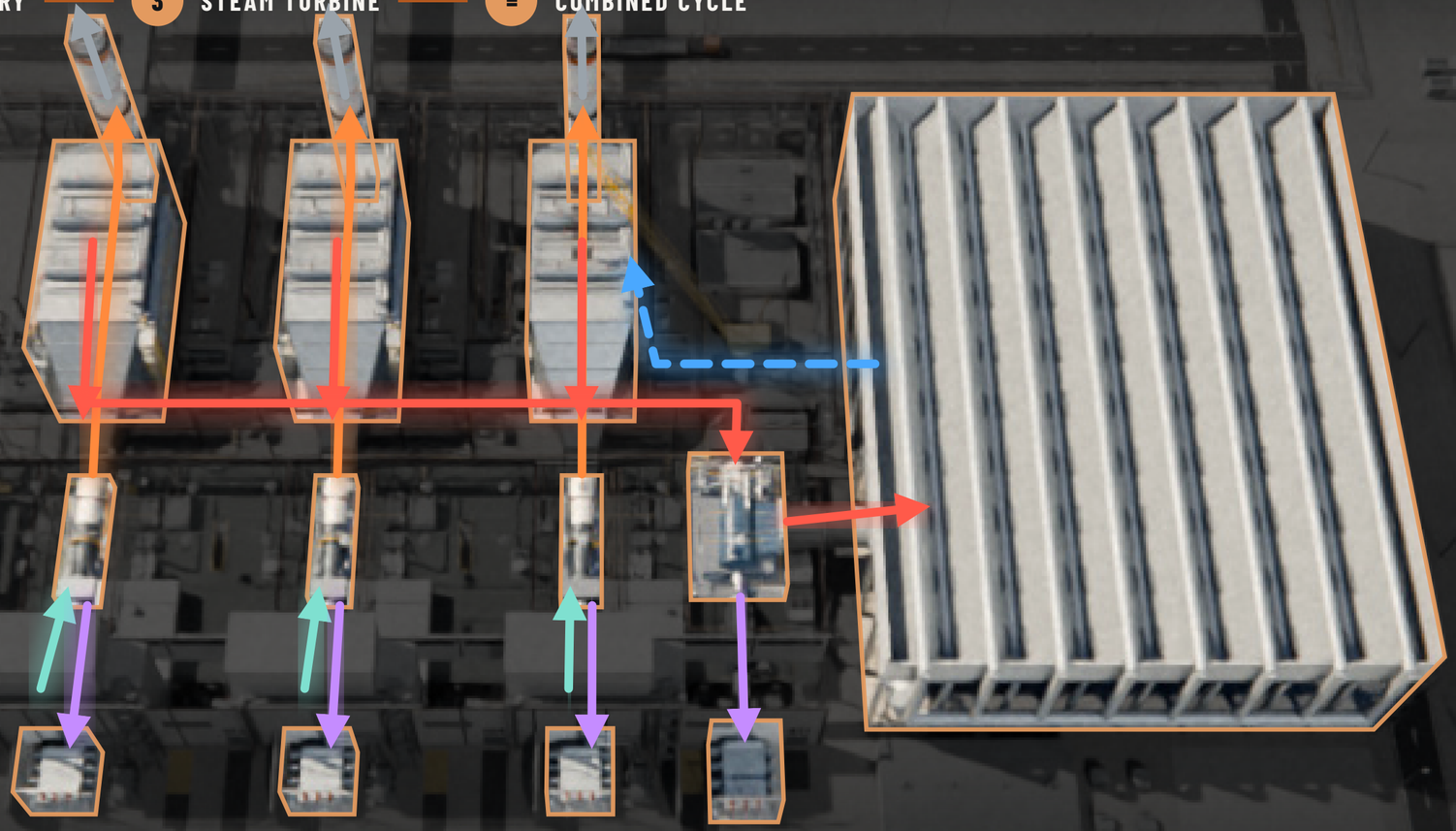
Steam from all three HRSGs spins one more turbine and generator: **about one-third more power.** The air-cooled condenser turns it back into water, and round it goes.

COMBINED CYCLE

BURN THE GAS ONCE, USE THE HEAT TWICE

EXHAUST · STEAM · WATER · POWER

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE — = COMBINED CYCLE



~60%
of the fuel becomes power, vs ~40% for a gas turbine alone

3 + 1
three gas turbines, three HRSGs, one steam turbine: a "3x1"

8,234
cables in this power block and its condenser, in our model

FUNDAMENTALS · THE BASICS

HOW AN HRSG WORKS

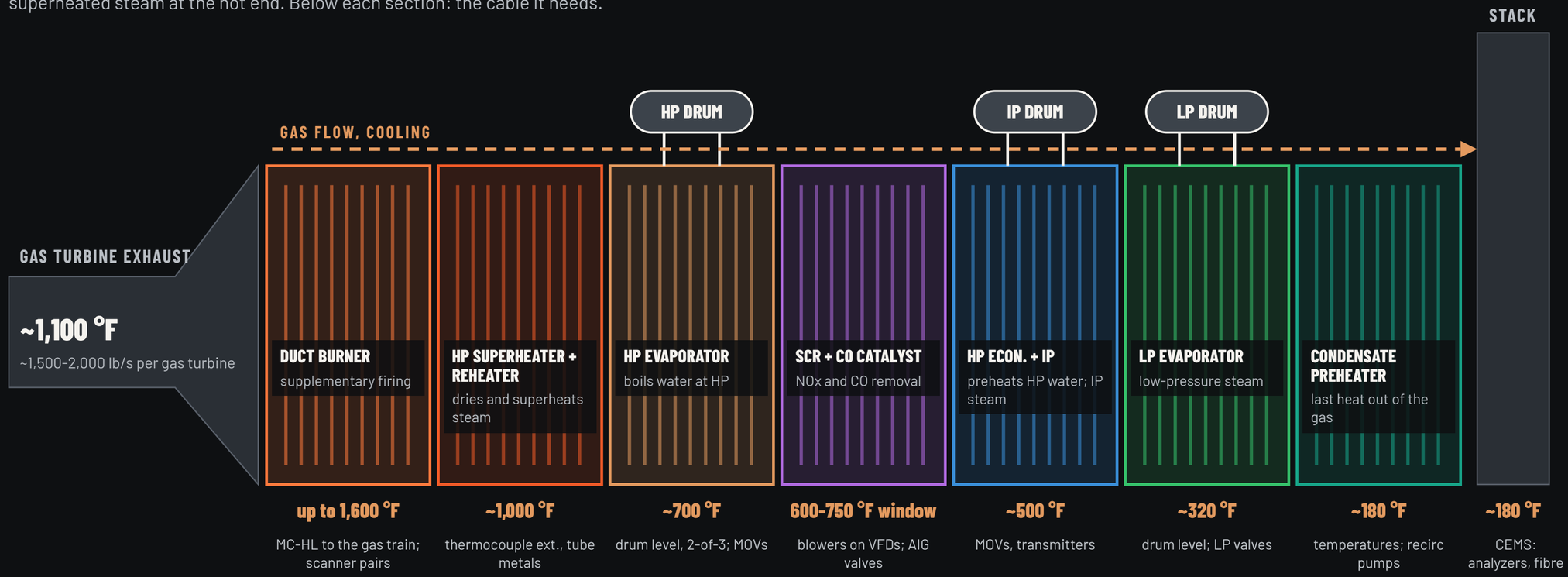
The exhaust crosses rows of finned tubes, hottest sections first; water enters at the cold end and leaves as superheated steam at the hot end. Below each section: the cable it needs.

WHAT IT IS

- **Without it:** ~60 % of the fuel's energy goes up the stack as ~1,100 °F exhaust (~40 % efficient).
- **With it:** a boiler heated by that exhaust; its steam drives the steam turbine.
- **The payoff:** ~1/3 more power from the same fuel (~60 %+ efficient).

FUN FACTS

- Fuel, CO2 and cost per MWh drop by about a third.
- The NOx catalyst (SCR) lives **inside** it, where the gas is just the right temperature.
- Duct burners can fire extra gas for peak demand.
- No HRSG: a gas-turbine plant. With one: combined cycle.



Steam to the steam turbine, typical: HP ~1,800-2,400 psig, ~1,050 °F · reheat / IP ~400-600 psig, ~1,050 °F · LP ~50-80 psig

~1/3
of a combined-cycle plant's output comes from HRSG steam, with no extra fuel (duct firing aside)

~60 %+
combined-cycle efficiency, against ~40 % for the gas turbine alone

3 x 3
three pressures, three drums; each drum level measured three ways, typically voted 2-out-of-3

~1 hour
a hot start: about an hour: every start cycles the tubes, which is why tube-metal thermocouples matter

Typical three-pressure reheat HRSG with SCR behind an H-class gas turbine; section order, temperatures and pressures vary by design.

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FROM THE GENERATOR TO THE LIGHT SWITCH

Power is made at medium voltage, raised to high voltage to travel, then stepped down again near the customer.

1

GENERATOR

13.8–25 kV

21 kV in SW-3X1

CABLE

isolated-phase bus, MV-105

›

2

STEP-UP (GSU)

→ 115–765 kV

230 kV in SW-3X1

CABLE

HV XLPE cable, switchyard bus

›

3

TRANSMISSION

115–765 kV

long distance, overhead

CABLE

overhead conductor, OPGW

›

4

SUBSTATION

→ 4–35 kV

steps it down for the town

CABLE

substation cable, grounding

›

5

DISTRIBUTION

4–35 kV

feeders along the streets

CABLE

MV URD, overhead distribution

›

6

CUSTOMER

480 / 240 / 120 V

factory, office, home

CABLE

building wire, MC, service entrance

← INSIDE THE PLANT: TODAY

Higher voltage, lower current, smaller losses: that is why transmission runs at hundreds of kV. Southwire has a product family at every step, from the generator to the outlet.

LEVELS, RATINGS AND THE CABLE THEY DRIVE

One HRSG train from the generator down to the field signals: each voltage level, the equipment and its rating, what it feeds, and the cable that follows.

LEVEL	EQUIPMENT · RATING	FEEDS · CURRENT	CABLE IT DRIVES
230 kV	**GSU-1, 21 / 230 kV step-up**	Plant to the grid, overhead line	**Overhead conductor (outside the HRSG scope)**
21 kV	**GT1 generator, isolated-phase bus; UAT-1 21 / 13.8 kV, 40 MVA**	UAT-1 secondary 1,674 A into SWGR-13.8-1	**4 sets of 3 x 1/C 750 kcmil MV-105**
13.8 kV	**SWGR-13.8-1: metal-clad, 1,200 A, 40 kA, arc-resistant**	Feed pumps BFP-1A / 1B, 5,000 kW each, direct-on-line, 232 A; T-R2A primary, 105 A	**3 x 1/C 250 kcmil MV-105 per pump; 3 x 1/C #2 to T-R2A**
480 V	**T-R2A 2,500 kVA → LC-R2A 4,000 A main, 65 kA → MCC-R2A-1 / -2, 800 A, 65 kA; VFDs**	Secondary 3,007 A; SCR blowers 2 x 110 kW (156 A); 45 HRSG MOVs; feed-pump and chemical-feed auxiliaries; welding receptacles	**10 sets of 4 x 1/C 500 kcmil; 3/0 TC-ER / MC-HL VFD; #10 TC-ER to each MOV**
120 V AC / 125 V DC	**UPS-R2A 10 kVA; 125 V DC battery, 60 cells; DC and UPS panels**	DCS, F&G node, breaker trip and close, burner-management power	**1/C #6 and control TC-ER 14 AWG**
Signals	**4-20 mA / HART, thermocouple millivolts, 24 V DC discretes**	Drum level, pressures, flame scanners, gas-path and tube-metal temperatures	**439 instrument + 84 thermocouple-extension cables, shielded pairs**

750 kW
connected at 480 V on one HRSG train, 78 circuits

2 x 5 MW
feed pumps: the largest motors on the train, at 13.8 kV

3,007 A
on the T-R2A secondary: 40 cables of 500 kcmil

74 %
of the train's cables are instrument or thermocouple pairs (68 % of the footage)

8.4 / 24.6
cal/cm^2 arc flash at 13.8 kV / 480 V (equipment labels)

Ratings are the model's typical values (Load List and nameplates), not engineered. Real plants vary: many put the feed pumps on a 4.16 or 6.9 kV bus.

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THE CABLE FAMILIES IN A POWER PLANT

From the biggest to the smallest. The whole SW-3X1 campus: 21,217 cables, 1,212 miles.

HV — 69–345 kV
XLPE: switchyard and gen-tie

MV — 5–35 kV
MV-105: big motors, transformers, feeders

LV POWER — 600 V
TC-ER, MC, ARMOR-X: motors, panels, lighting

VFD — 600 V–2 kV
drive-rated: fans and pumps on drives

CONTROL — 600 V
TC-ER multi-conductor: breakers, valves, interlocks

INSTRUMENTATION — 300 V
PLTC / ITC shielded pairs, thermocouple extension

FIBRE — –
controls network, protection

GROUNDING — bare Cu
the ground grid under everything

By length: Instrumentation 44% · LV 600 V power 20% · Control 18% · Fibre, fire alarm, other 11% · MV / HV 7%. The big cables get the attention; the small ones make up most of the miles.

Shares by footage from the SW-3X1 model cable schedule; voltage ratings typical.

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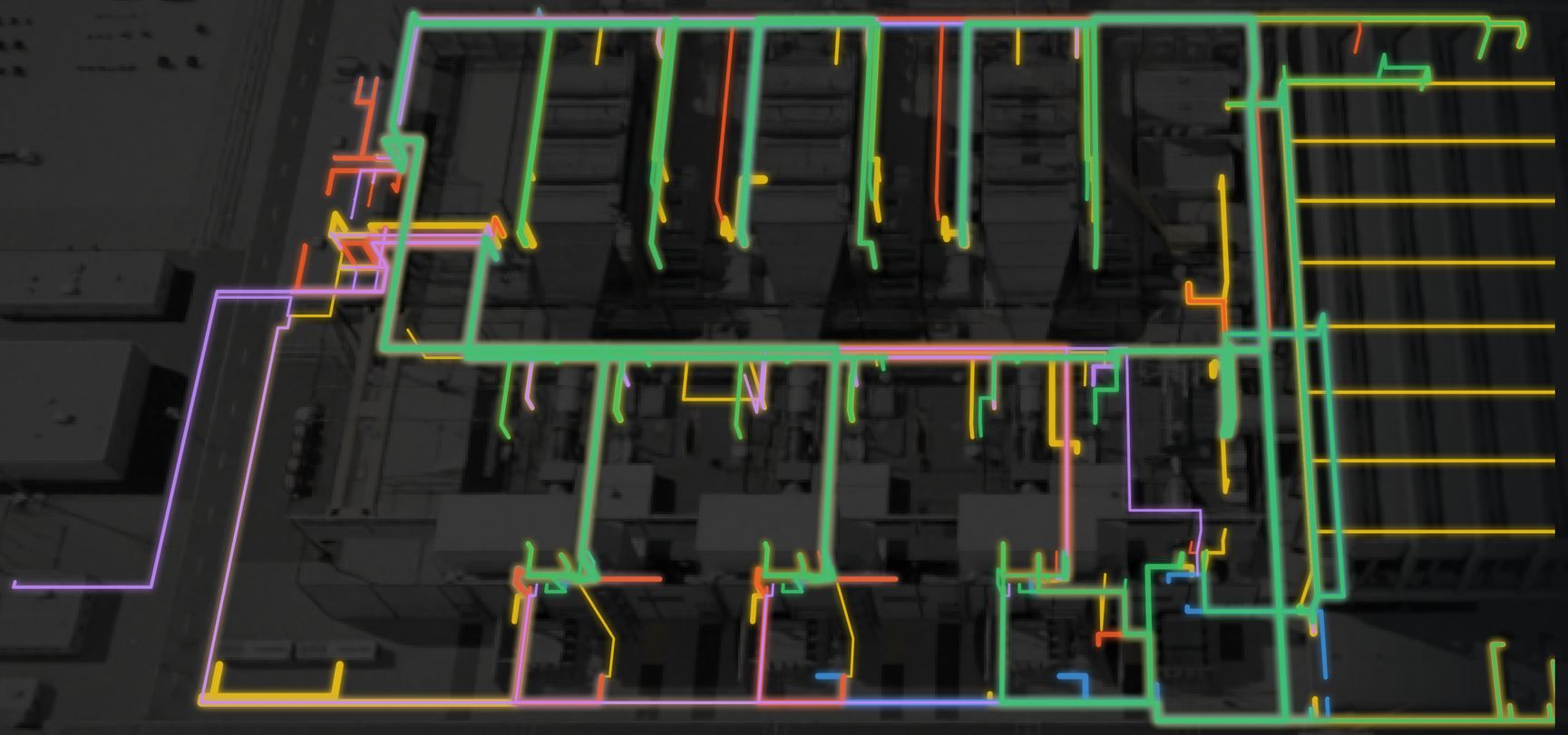
TAKE THE PLANT AWAY: THE CABLE STAYS



THE CABLE IN THIS PICTURE

8,234 cables · ~439 miles

MV / HV POWER — 232 · 28 mi
The big feeders: generators, transformers, the largest motors. MV-105, HV XLPE.

LV POWER — 1,160 · 95 mi
600 V to motors, valves, drives, lighting. TC-ER, ARMOR-X MC-HL, VFD cable.

CONTROL — 1,255 · 94 mi
Starts, stops and trips: breakers, interlocks, MCC starters. Multi-conductor TC-ER.

INSTRUMENTATION — 5,484 · 202 mi
The plant's nerves: pressure, flow, temperature. Shielded pairs, thermocouple extension.

FIBRE AND FIRE ALARM — 103 · 21 mi
The control network, protection and life safety.

The power block and its air-cooled condenser only, from the SW-3X1 model cable schedule (building wire, grounding and overhead not counted). Conceptual estimate, not engineered.

Every line is a route from the model's cable schedule, in trays, conduit and duct banks; thicker lines carry more cables. **The power block is a web of copper.**

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WHAT A POWER PLANT DOES TO A CABLE

STRESS	WHERE	THE ANSWER
HEAT	beside turbines, HRSGs and steam lines	high-temperature ratings, route away from hot surfaces
CYCLING	daily starts: heat, vibration, moving terminations	flexible strands, good terminations, support
ELECTRICAL NOISE	variable-frequency drives on fans and pumps	VFD-rated, shielded cable
HAZARDOUS AREAS	gas skids and fuel yards (Class I, Div 2)	ARMOR-X MC-HL, no conduit
FIRE AND SMOKE	halls, control rooms, cable spreading rooms	flame-retardant, low-smoke, circuit integrity
OUTDOORS	sun, oil, water, burial	sunlight- and oil-resistant jackets, direct-burial ratings
SCHEDULE	long pulls on the critical path	SIMpull reels, cut-to-length, pre-terminated

The wrong cable fails early, and a cable failure can trip a unit. That is why the spec is strict, and why it is written for each area of the plant.

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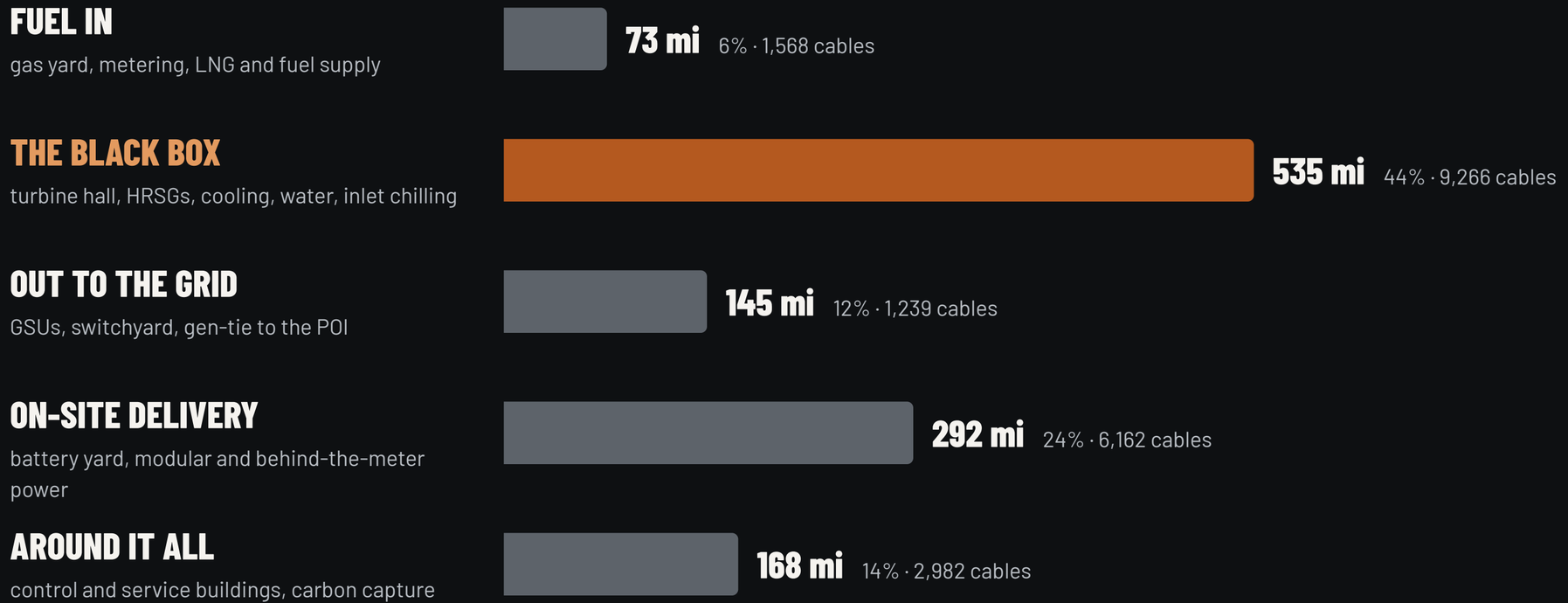
FOLLOW THE POWER: THE CABLE IS EVERYWHERE

THE WHOLE SW-3X1 CAMPUS BY PLANT AREA, FROM THE MODEL'S CABLE SCHEDULE (1,212 MILES)



THE POI

Point of interconnection: where the plant meets the grid. The plant side is the owner's and EPC's scope; past it, the utility or transmission owner builds and buys.

The gen-tie (often HV underground) runs from the switchyard to the POI, and the scope line there is often blurred: a place we help.

Most of the cable is inside the box: thousands of instrument, control and motor circuits in the turbine hall and HRSGs. The grid connection carries the biggest cables; the power block carries the most.

SAME PHYSICS, DIFFERENT OWNERS

The owner decides how the project is structured, and how it gets bought. An independent power producer (IPP) builds each plant as its own company, sells the power on contract or into a market, and funds it with investors.

THE STRUCTURAL EQUATION

THERMAL IPP = **PROJECTCO** (one company per plant) + **ASSET** (conversion plant) + **CONTRACT** (power purchase agreement) + **CUSTOMER** (off-taker / load) + **CAPITAL** (investors, tax equity) + **MARKET** (ERCOT, PJM, MISO)

POWER OWNER ARCHETYPES

IOUS / CO-OPS — REGULATED

- Utility-led, rate-based ownership
- Own and operate generation
- Buy like utilities: standards, cycles

HYBRID IPPS — MIXED MODEL

- Utility-affiliated IPP subsidiaries
- Bridge regulated and competitive
- Two buying behaviours

PURE IPPS — COMPETITIVE

- Privately owned, investor-backed projects
- A direct fit with the project equation
- Speed and cost drive decisions

OUR SALES FOCUS FOR COMPETITIVE THERMAL POWER

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GREENFIELD AND BROWNFIELD: FIVE KINDS OF SITE

GREENFIELD · NEW SITES 137 GW in the register

77.0 GW

GREENFIELD, GRID-CONNECTED

A new plant on open land, tied to the grid.

Everything is new: the full cable schedule, the switchyard and an HV gen-tie to the POI.

54.5 GW

DATA-CENTRE / BEHIND-THE-METER CAMPUS

Power built next to the load it serves.

Modular engines, turbines and batteries; MV collection to the campus; fast schedules, bought by package builders.

BROWNFIELD · EXISTING SITES 69 GW in the register

41.3 GW

COAL-SITE REPOWER

New gas units on a retired coal or boiler site.

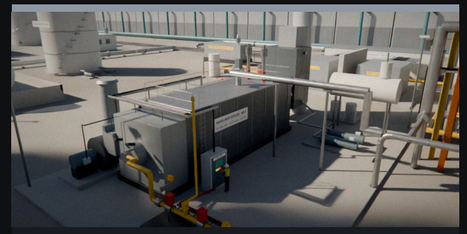
Reuses the land, water and grid connection; new plant cable tied into an old switchyard.

21.4 GW

EXPANSION AT AN OPERATING PLANT

New units beside ones that keep running.

Work around live equipment; tie-ins wait for outages; shared switchyard and controls.



11.6 GW

INDUSTRIAL CAPTIVE AND SITE REUSE

Power, often with steam, for a factory or refinery.

Hazardous areas (MC-HL), tight routes, the host plant's own standards.

And on plants already running: outages, HRSG module replacements, upgrades and conversions. There the outage clock is the cost, so cable is staged, pre-terminated and pulled fast. Utilities mostly reuse old sites; IPPs and behind-the-meter developers mostly build new.

Gross GW by site class, Southwire PowerGen register, Sep 11, 2026 (205.8 GW eligible, COD 2026–2035). Site classes provisional (Rev01).

ONE COMPANY SPECIFIES THE CABLE, ANOTHER ONE BUYS IT

DEVELOP

site, permits, interconnection request, fuel and water

WHO
Owner / developer

YEARS: THE INTERCONNECTION QUEUE IS OFTEN THE LONG POLE

FINANCE

PPA or rate case, turbine slot, financing decision (FID)

WHO
Owner, lenders, OEM

TURBINES ARE RESERVED YEARS AHEAD

ENGINEER

owner's engineer writes the specs; the EPC does detailed design

WHO
Owner's engineer, EPC

CABLE IS SPECIFIED HERE

PROCURE

EPC and package builders buy equipment and bulk materials

WHO
EPC, OEMs, package builders

CABLE IS BOUGHT HERE

BUILD

civil, mechanical, then electrical: trays, pulls, terminations

WHO
EPC and electrical contractor

CABLE PULLS SIT ON THE CRITICAL PATH

OPERATE

commercial operation (COD), outages, upgrades, repowering

WHO
Owner / operator

REPLACEMENT AND BROWNFIELD WORK

The owner pays, the owner's engineer writes the spec, the EPC or package builder places the order. Be in the spec early, and be ready when the EPC buys: by then the schedule is tight and cable is on the critical path.

A MODERN GAS PLANT IS MORE THAN THE POWER BLOCK

PRIVATE 5G / LTE

A monopole with massive-MIMO antennas: the plant's own wireless network.

CABLE

fibre backhaul, tower power, grounding

EV CHARGING + SOLAR CARPORT

Chargers at every bay, a DC fast charger, PV on the carport roof.

CABLE

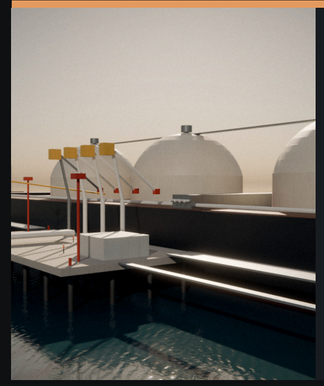
LV feeders, PV wire, a pad-mount transformer

BEHIND-THE-METER DATA CENTRE

A ~150 MW campus fed straight from the plant and the modular yard.

CABLE

34.5 kV collection, cable bus, UPS feeders



LNG BY SEA

A marine terminal: the carrier at the berth, unloading arms, the tank onshore.

CABLE

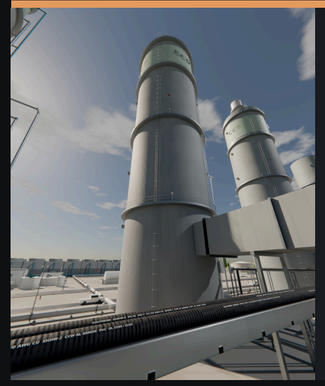
hazardous-area cable, long runs to the jetty, MV pumps

HYDROGEN

Tube-trailer bay and receiving station; an electrolyzer building beside it.

CABLE

Class I Div 2 runs, gas detection, DC cable



CARBON CAPTURE

Absorber, stripper and compressors on the stack exhaust.

CABLE

a second plant's motors, drives and instruments

All six are in the SW-3X1 model. Each one adds its own network of cable on top of the power block: the campus is the customer, not just the turbines.

THE CAMPUS · FROM THE SEA TO THE GRID

ONE SITE, SEVEN STOPS

1. GRID CONNECTION AND SWITCHYARD
2. TURBINE HALL
3. HRSG AND ITS E-HOUSE
4. COOLING AND WATER
5. MODULAR GENERATION
6. BEHIND THE METER: THE DATA CENTRE
7. FUEL: GAS YARD AND LNG

LNG arrives by ship at the marine terminal; the power leaves on 230 kV lines. Everything in between runs on cable.

TOUR STOP 1 OF 7

1 GRID CONNECTION AND SWITCHYARD

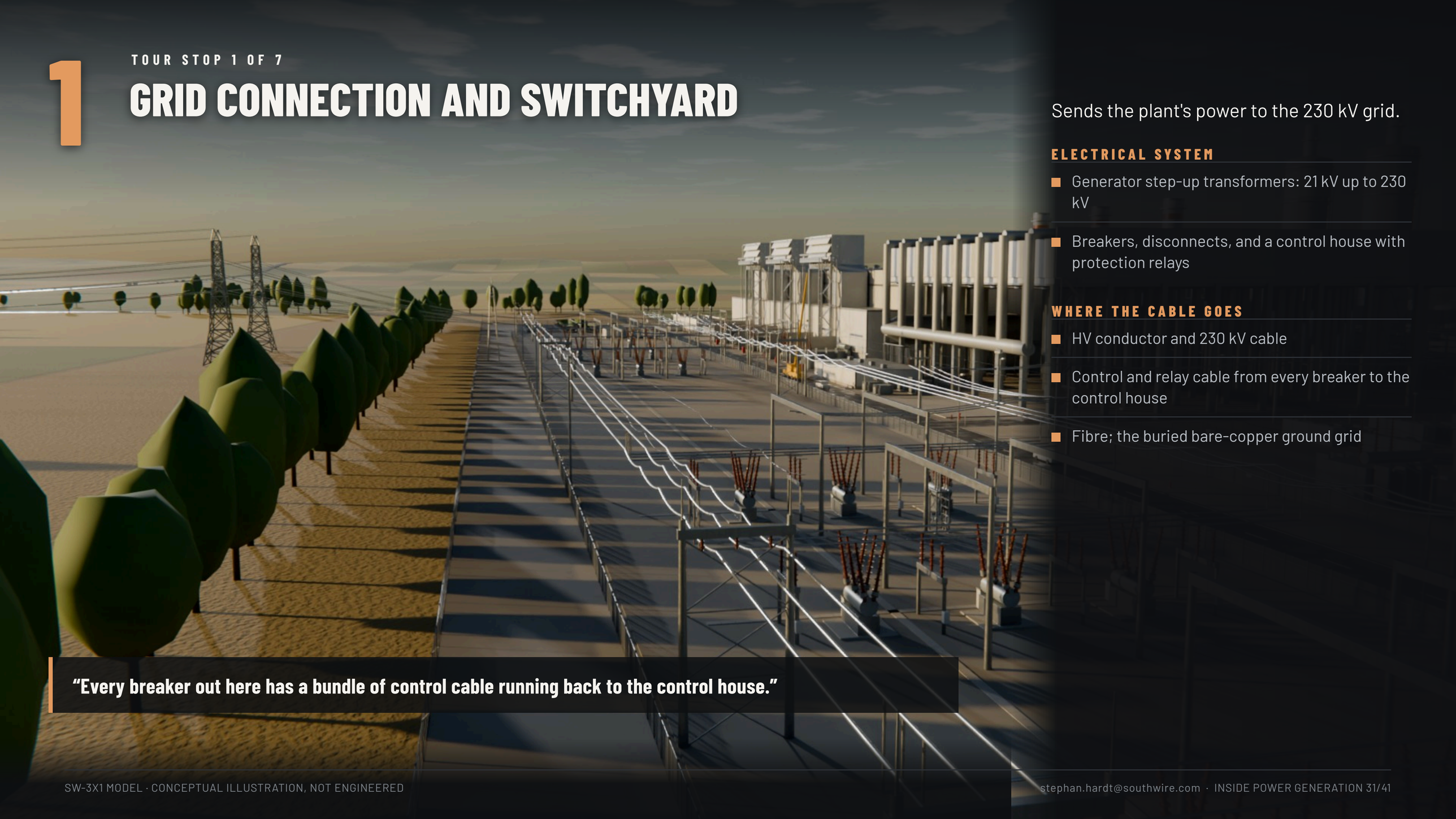
Sends the plant's power to the 230 kV grid.

ELECTRICAL SYSTEM

- Generator step-up transformers: 21 kV up to 230 kV
- Breakers, disconnects, and a control house with protection relays

WHERE THE CABLE GOES

- HV conductor and 230 kV cable
- Control and relay cable from every breaker to the control house
- Fibre; the buried bare-copper ground grid



“Every breaker out here has a bundle of control cable running back to the control house.”

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED

TOUR STOP 2 OF 7

2 TURBINE HALL

Three H-class gas turbines and one steam turbine turn the generators.

ELECTRICAL SYSTEM

- Isolated-phase bus from the generators to the step-up transformers
- Unit auxiliary transformers feed the 13.8 kV switchgear
- Static starters spin the gas turbines up

WHERE THE CABLE GOES

- MV-105 to big motors and transformers
- TC-ER power and control to the lube-oil and hydraulic skids
- Thousands of instrument pairs: vibration, temperatures

“The machines make the power; the cable keeps them running and tells us how they feel.”

TOUR STOP 3 OF 7

3 HRSG AND ITS E-HOUSE

Turns turbine exhaust heat into steam; the SCR inside cuts NOx.

ELECTRICAL SYSTEM

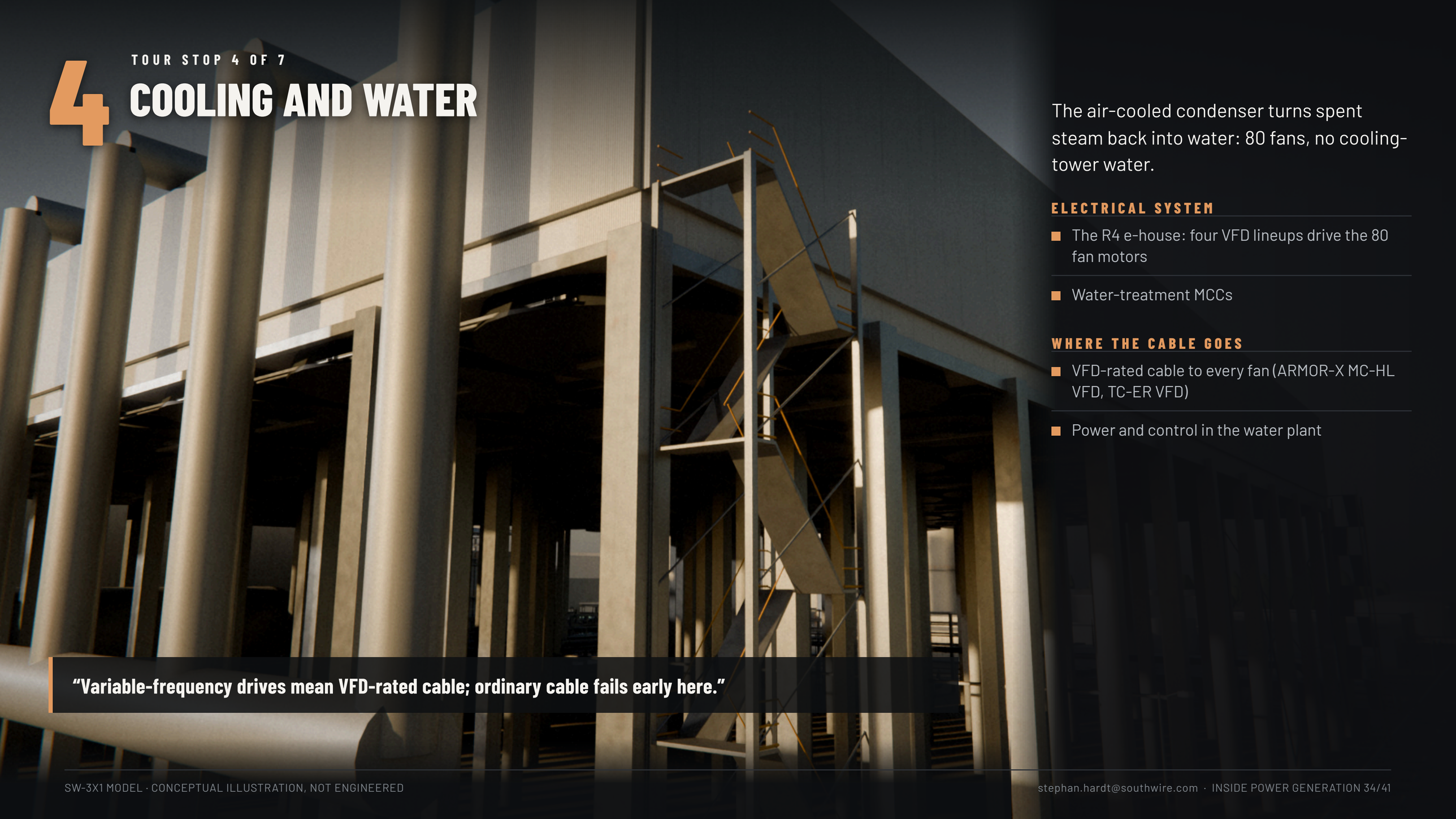
- Two 13.8 kV feed pumps
- A transformer feeds the e-house: 480 V switchgear, MCCs, drives, DCS
- CEMS shelter at the stack

WHERE THE CABLE GOES

- One HRSG train: 707 cables, about 157,000 ft
- About three quarters are instrument and thermocouple pairs
- ARMOR-X MC-HL at the duct-burner gas skid

“One boiler train is about 30 miles of cable, and most of it is small instrument cable.”

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED



TOUR STOP 4 OF 7

4 COOLING AND WATER

The air-cooled condenser turns spent steam back into water: 80 fans, no cooling-tower water.

ELECTRICAL SYSTEM

- The R4 e-house: four VFD lineups drive the 80 fan motors
- Water-treatment MCCs

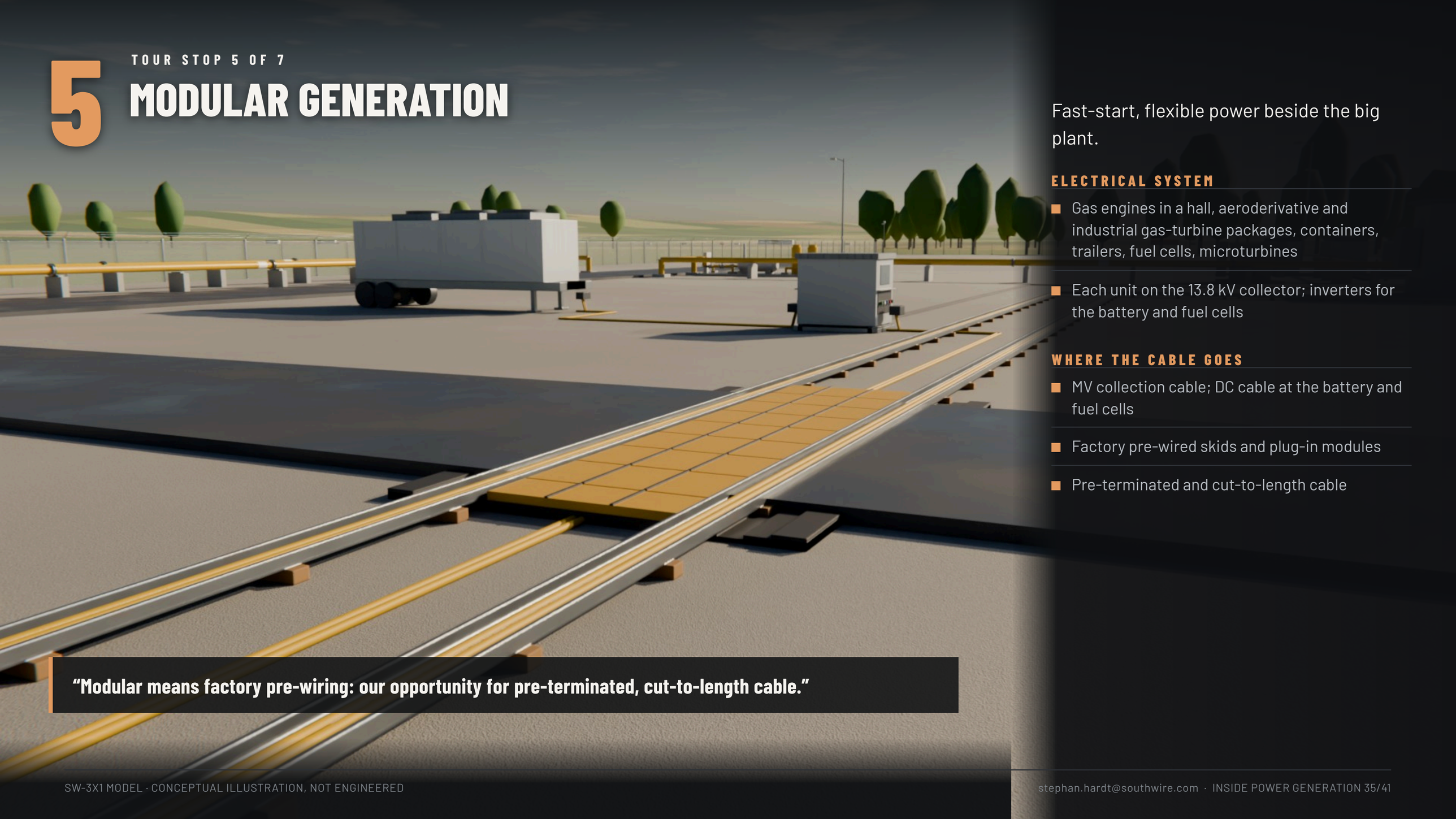
WHERE THE CABLE GOES

- VFD-rated cable to every fan (ARMOR-X MC-HL VFD, TC-ER VFD)
- Power and control in the water plant

“Variable-frequency drives mean VFD-rated cable; ordinary cable fails early here.”

TOUR STOP 5 OF 7

5 MODULAR GENERATION



Fast-start, flexible power beside the big plant.

ELECTRICAL SYSTEM

- Gas engines in a hall, aeroderivative and industrial gas-turbine packages, containers, trailers, fuel cells, microturbines
- Each unit on the 13.8 kV collector; inverters for the battery and fuel cells

WHERE THE CABLE GOES

- MV collection cable; DC cable at the battery and fuel cells
- Factory pre-wired skids and plug-in modules
- Pre-terminated and cut-to-length cable

“Modular means factory pre-wiring: our opportunity for pre-terminated, cut-to-length cable.”

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED

TOUR STOP 6 OF 7

6 BEHIND THE METER: THE DATA CENTRE

A data-centre campus powered directly by the plant's modular units, with the grid as backup.

ELECTRICAL SYSTEM

- 13.8 to 34.5 kV step-up transformers and double-bus switchgear
- A 40 MW battery absorbs load steps
- Two data halls fed 2N through pad-mount transformers and UPS

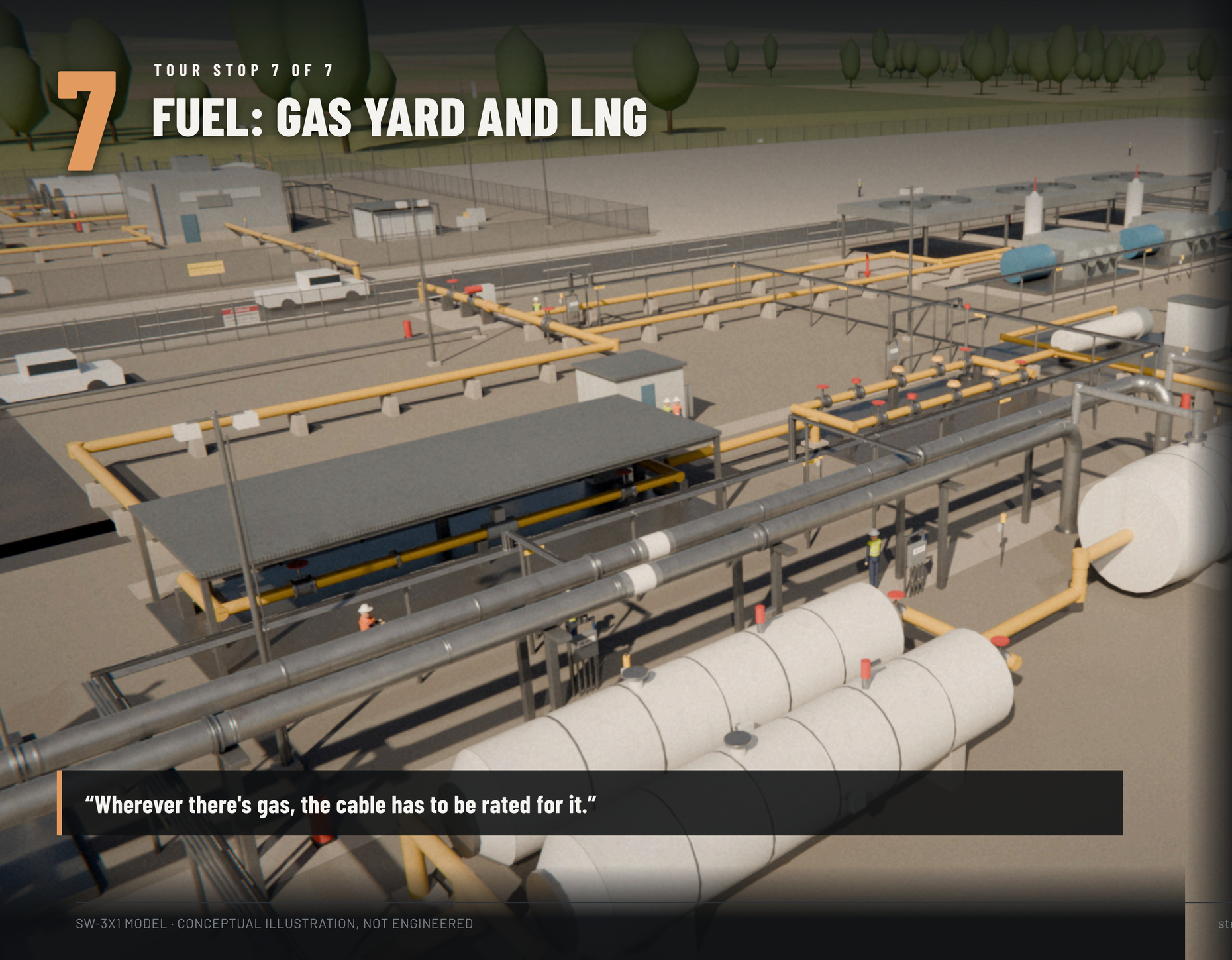
WHERE THE CABLE GOES

- 35 kV MV cable
- Large LV feeders to the UPS skids
- Fibre everywhere

“Data centres are why gas is growing again, and they are cable-dense on both the A and the B side.”

TOUR STOP 7 OF 7

7 FUEL: GAS YARD AND LNG



Brings the fuel in and conditions it; an LNG satellite holds about three hours of backup fuel.

ELECTRICAL SYSTEM

- Metering and regulating station; the LNG e-house (13.8 kV / 480 V)
- Pumps, vaporizers, boil-off compressor

WHERE THE CABLE GOES

- ARMOR-X MC-HL for Class I Div 1 and 2
- TC-ER control, PLTC instrumentation and fire alarm on one strut rack

> **“Wherever there's gas, the cable has to be rated for it.”**

THE WHOLE CAMPUS IN ONE NUMBER

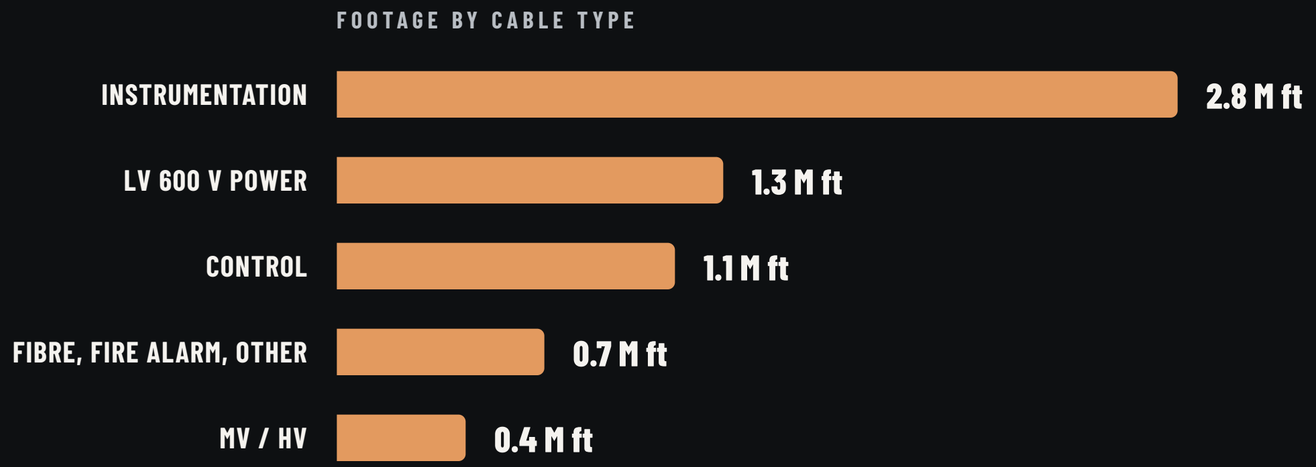
6.4M

FEET OF CABLE · ABOUT 1,212 MILES

6,011
CABLE RUNS

21,217
CABLES

From the model's cable schedule: every circuit on the campus, from the 230 kV switchyard to the last thermocouple.



Most of the footage is small cable: instrument and control. MV is the high-value part, but it isn't the volume.

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WHAT THE SPEC CALLS FOR

Every cable application on the HRSG, the requirements an owner's or EPC's cable specification writes for it, and the Southwire product built to meet them.

APPLICATION · WHERE	WHAT THE SPECIFICATION CALLS FOR	SOUTHWIRE
MV POWER, 15 KV FEED-PUMP MOTORS; THE T-R2A PRIMARY	1/C Cu, EPR insulation, **133 % level**, 90 °C, tape shield, PVC jacket. ICEA S-93-639, UL 1072 (MV-105). Terminations IEEE 48; acceptance test VLF (IEEE 400.2) / NETA ATS	**MV-105 15 kV, 220 mil NL-EPR 133 %, SIMpull · SPEC 46302**
480 V FEEDERS AND POWER T-R2A SECONDARY, LC-R2A TO MCCS, MOTORS AND MOVS	Cu XHHW-2, 90 °C wet / dry, sized at **125 %** of full load; voltage drop ≤ 3 %. ICEA S-95-658, UL 44 / UL 1277; 10+ parallel sets: compare with bus duct	**XHHW-2 1/C (SIMpull) · Type TC-ER 3/C + ground · SPEC 45252**
TRAY CABLE: POWER AND CONTROL EVERY TRAY BETWEEN THE E-HOUSE AND THE HRSG	Type TC-ER (UL 1277) so it may leave the tray as an exposed run to the equipment (NEC 336.10); sunlight resistant, direct burial; **IEEE 1202 / UL 1685** 70,000 Btu/h vertical-tray flame test	**Control TC-ER 14 AWG multiconductor · SPEC 45053**
HAZARDOUS AREA: GAS TRAIN, DUCT BURNER DUCT-BURNER FUEL SKID, BMS PANEL TO THE VALVES	Class I Div 2 wiring methods per NEC 501.10(B); MC-HL (UL 1569 + UL 2225) also qualifies for Div 1. Burner management per **NFPA 85**, duct burner UL 295, listed fittings	**ARMOR-X MC-HL, welded armor · SPEC 45352**
VARIABLE-FREQUENCY DRIVES SCR DILUTION-AIR BLOWERS, FD FANS	VFD-rated: symmetrical three-ground construction, overall shield or continuous armor for the common-mode current, XLPE insulation for reflected-wave peaks	**ARMOR-X MC-HL VFD · TC-ER VFD · SPEC 45352**
INSTRUMENTATION, 4-20 MA / HART DRUM LEVEL, PRESSURES, FLAME SCANNERS, ANALYZERS	**Individually shielded** twisted pairs + overall shield, drain wires; 300 V PLTC (UL 13, NEC 725) or ITC; ICEA S-73-532; segregated from power in tray	**Instrumentation PLTC, SPOS · SPEC 43303**
THERMOCOUPLE EXTENSION GAS-PATH, TUBE-METAL AND DUCT TEMPERATURES	Alloy matched to the thermocouple (type K = KX), ANSI MC96.1 / ASTM E230 colour code; **never copper**; shielded pairs; PLTC; 105 °C near the casing	**Thermocouple extension, PLTC multi-pair · SPEC 42345 / 42350**
FIRE ALARM, FIBRE, BUILDING WIRE F&G LOOPS, CEMS DATA, E-HOUSE LIGHTING	FPLR / FPLP (NEC 760, NFPA 72); single-mode fibre, armored in tray; THHN / THWN-2 in conduit (UL 83)	**SIMpull THHN / THWN-2 · fire-alarm and fibre cable**

CHECKED AGAINST A REAL PLANT

A recent US utility combined-cycle filing shows the same structure: **HRSG MCCs A / B** fed from 480 V switchgear through 2,500 / 3,300 kVA transformers off the auxiliary bus (6.9 kV there, 13.8 kV here).

DESIGN AND INSTALL

Raceway and cable design for generating stations: IEEE 422. Grounding: IEEE 80 / 665. Area classification: NEC 500, API RP 500. Field acceptance: NETA ATS.

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PAIN POINTS AND THE ANSWER

TECHNICAL PAIN POINTS

- **MOSTLY SIGNAL CABLE** ~74 % of an HRSG's cables are instrument or thermocouple pairs; bad shield grounding or routing next to drives and MV makes noisy drum-level and flame-scanner signals that can trip the unit
- **THERMOCOUPLE EXTENSION ERRORS** ~84 runs per train: wrong alloy, reversed polarity or copper substituted; found as a bad tube-metal reading after startup
- **HEAT AND SUN** Hot casings, roof penthouses, drum platforms: 90-105 °C ratings, sunlight-resistant jackets, derating
- **MOVEMENT AND CYCLING** The HRSG grows inches when hot and cycles daily: slack at module joints, vibration, termination fatigue
- **GAS IN THE BOILER** Duct burners and fuel skids make Class I Div 2 areas; NFPA 85 burner management; conversions add it
- **BIG PARALLEL FEEDERS** 3,007 A transformer secondary = 40 cables of 500 kcmil; equal lengths in congested tray
- **DRIVES** SCR blowers and fans on VFDs: bearing currents and voltage spikes without VFD-rated cable
- **SCOPE GAPS** Vendor junction boxes, EPC home runs: legs fall between the two (our own schedule missed the 480 V secondaries)
- **SCHEDULE AND LABOUR** Cable is on the critical path, above all in an outage; electricians are scarce; no room to stage reels

VALUE PROPOSITION

- **ONE SUPPLIER, WHOLE SCOPE** MV-105, TC-ER, ARMOR-X MC-HL and VFD, instrumentation, thermocouple extension, THHN, fire alarm: one PO, one spec-compliance package (IEEE 1202, UL 1277 / 2225 / 13)
- **FASTER PULLS: SIMPULL** Pre-lubricated jacket cuts pulling force and time; parallel conductors on one reel go in one pull
- **STAGED FOR THE OUTAGE** Cut-to-length reels labelled per pull, SIMpull Cradles for one-person payoff: ready before the trip
- **ONE CABLE FOR HAZARDOUS AREAS** ARMOR-X MC-HL is rated Div 2 and Div 1: fewer conduits and seals, unaffected by reclassification
- **THE RIGHT CONSTRUCTION** VFD-rated cable for drives, individually shielded pairs, matched-alloy colour-coded thermocouple extension
- **ENGINEERING UP FRONT** Ampacity, voltage drop, spec review; takeoff ready: ~707 cables, ~157,000 ft per train, ~472,000 ft for three
- **SUSTAINABILITY** Reel returns and scrap recycling for cable cut back in brownfield outages

"On an HRSG, cable is three-quarters instrumentation and fully on the critical path. One spec-compliant supply, staged and fast to pull: wired on time, reading right the first time."

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THREE THINGS TO REMEMBER

TAKE AWAY

1 **GAS IS GROWING AGAIN**

driven by data centres and the need for firm power.

2 **MOST OF A PLANT'S CABLE IS SMALL**

instrument and control; the big MV and power cable is the smaller share.

3 **SOUTHWIRE COVERS THE WHOLE PLANT**

one supplier, staged on SIMpull reels, spec-compliant from MV down to the thermocouple.

Southwire®
POWER GENERATION SOLUTIONS

Q&A

Questions from the chat. The recording and the 3D viewer link will be shared after the session.

Stephan Hardt
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